

The operator norm of matrix truncations

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1 Introduction

An interesting question showed up the other day. First, consider a Hamiltonian $H = H_0 + \epsilon V$ where H_0 is degeneracy-free, and a transformation $U = e^T$ diagonalizing H , i.e. such that $e^{-T} H e^T$ is diagonal in the eigenbasis of H_0 . Let's write T as a power series: $T = \sum_{q=1}^{\infty} \epsilon^q T_q$. Let $\mathcal{T}$ denote the superoperator $[T, \cdot]$. Then we can expand $e^{-T} H e^T = e^{\mathcal{T}} H$ in powers of ϵ :

$$e^{\mathcal{T}} H = H_0 + \sum_{q=1}^{\infty} \epsilon^q \mathcal{T}_q H_0 + V^{(q-1)} \quad (1)$$

where $V^{(q-1)}$ is Hermitian and a function of $T_{k < q}$ and V (the specific form is not important). Clearly, if we choose T_q such that $\mathcal{T}_q H_0 = -O(V^{(q-1)})$, where $O(V^{(q-1)})$ are the off-diagonal elements of $V^{(q-1)}$, then the transformed Hamiltonian will be diagonal at all orders.

At this point, we define the superoperator $\mathbb{P}^+$ which projects onto the lower-left-triangular part of a matrix, so that if A is an operator, $\mathbb{P}^+ A$ strictly raises the eigenvalue of H_0 . Then we define the integral operator

$$I(A) = \int_0^{\infty} e^{-H_0 s} \mathbb{P}^+ A e^{H_0 s} \quad (2)$$

Notice that the norm of $I(A)$ satisfies

$$\|I(A)\| \leq \|\mathbb{P}^+ A\| \int_0^{\infty} ds e^{-\Delta s} = \frac{\|\mathbb{P}^+ A\|}{\Delta} \quad (3)$$

where Δ is the minimum gap in the spectrum. We then notice that $I(A)$ satisfies the following property:

$$[H_0, I(A)] = \int_0^{\infty} ds [H_0, e^{-H_0 s} \mathbb{P}^+ A e^{H_0 s}] = \sum_{j>i} (E_j - E_i) \langle j | A | i \rangle \int_0^{\infty} ds e^{-(E_j - E_i)s} = \sum_{j>i} \langle j | A | i \rangle \quad (4)$$

Thus, defining $L(A) \equiv I(A) - [I(A)]^\dagger$, if A is Hermitian, then

$$[H_0, L(A)] = O(A) \quad (5)$$

This is exactly what we wanted, and putting $T_q = L(V^{(q-1)})$, then if the series converges, it diagonalizes the Hamiltonian. So when does this series converge? In order to answer that, we need to estimate $\|\mathbb{P}^+ A\|$.

2 Matrix truncation as averaging

In searching for an answer to this question, I came across a great article by Bhatia [1] (the same person who wrote the matrix bible [2]). Suppose that $\|\cdot\|$ is a unitary-invariant norm. The easiest way to estimate the norm of a super-operator is to describe it as an average over some set of unitary matrices. Let $U_\theta = \text{diag}(1, e^{-i\theta}, e^{-2i\theta}, \dots, e^{-di\theta})$, where d is the dimension of the space. Then we notice that $(U^\dagger A U)_{ij} = e^{(i-j)\theta} A_{ij}$. Therefore the superoperator

$$A \mapsto \frac{1}{2\pi} \int_0^{2\pi} d\theta U_\theta^\dagger A U_\theta e^{ik\theta} \quad (6)$$

projects A onto the k^{th} diagonal, where $i - j = k$. Thus we can write the superoperator $\mathbb{P}^+$ as

$$\mathbb{P}^+ A = \frac{1}{2\pi} \sum_{k=1}^d \int_0^{2\pi} d\theta U_\theta^\dagger A U_\theta e^{ik\theta} \quad (7)$$

Estimating the norm and using unitary invariance gives

$$\|\mathbb{P}^+ A\| = \frac{\|A\|}{2\pi} \int_0^{2\pi} d\theta \left| \sum_{k=1}^d e^{ik\theta} \right| \quad (8)$$

This looks very much like the L_1 norm of the Dirichlet kernel, and we can use a very similar strategy to bound its norm. First we have

$$\left| \sum_{k=1}^d e^{ik\theta} \right| = \left| \frac{e^{i\theta} - e^{di\theta}}{1 - e^{i\theta}} \right| = \left| -\frac{2ie^{i\theta/2}(1 - e^{(d-1)i\theta})}{\sin(\theta/2)} \right| = \left| \frac{4\sin^2([d-1]\theta/2)}{\sin(\theta/2)} \right| \quad (9)$$

On the interval $[0, \pi/(d-1)]$, we have

$$\frac{|\sin^2([d-1]\theta/2)|}{|\sin(\theta/2)|} \leq (d-1)^2\theta \quad (10)$$

and on the interval $[\frac{\pi}{d-1}, \pi - \frac{\pi}{d-1})$ we have

$$\frac{|\sin^2([d-1]\theta/2)|}{|\sin(\theta/2)|} \leq \frac{4}{\theta} \quad (11)$$

Integrating these, we have

$$\int_0^{\frac{\pi}{d-1}} (d-1)^2\theta d\theta + \int_{\frac{\pi}{d-1}}^{\pi - \frac{\pi}{d-1}} \frac{4}{\theta} d\theta \leq \frac{\pi}{2} + 4\ln(d-2) \quad (12)$$

If we repeat this argument for the intervals $[\pi - \pi/(d-1), \pi + \pi/(d-1)]$, $[\pi + \pi/(d-1), 2\pi - \pi/(d-1)]$, and $[2\pi - \pi/(d-1), 2\pi]$, we obtain

$$\int_0^{2\pi} d\theta \frac{\sin^2([d-1]\theta/2)}{\sin(\theta/2)} \leq 2\pi + 8\ln(d-2) \quad (13)$$

Taken altogether, we have

$$\|\mathbb{P}^+ A\| \leq \frac{\|A\|}{\pi} [2 + 32\ln(d-2)] \leq 12\|A\| \ln(d) \quad (14)$$

where the last inequality holds when $d > 1$. Therefore we have a bound that grows very slowly with d , but is in fact unbounded as $d \rightarrow \infty$. The prefactor is not tight; it can be made as small as $\frac{1}{\pi}$ for large d . It turns out that the form of the bound is tight; the Hilbert matrix, with entries $H_{ij} = (i - j)^{-1}$, has a norm bounded by π , but the norm of its lower-triangular truncation grows as $\frac{4}{5} \ln(d)$.

For some applications, this is already good enough. In particular, if our system consists of a finite number of sites with a finite onsite Hilbert space dimension, then this bound grows linearly in the number of sites.

References

- [1] Rajendra Bhatia. Pinching, trimming, truncating, and averaging of matrices. *The American Mathematical Monthly*, 107(7):602–608, 2000.
- [2] Rajendra Bhatia. *Matrix analysis*, volume 169. Springer Science & Business Media, 2013.